

2.1 Pen and Paper

Epipolar Geometry

$$a) [t_1]_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \quad \tilde{E}_1 = [t_1]_x R = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \quad \tilde{l}_1 = \tilde{E}_1^T \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ y \\ -x \end{pmatrix} \quad \tilde{l}_2 = \tilde{E}_1^T \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ y \\ -x \end{pmatrix}$$

$$[t_2]_x = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \quad \tilde{E}_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \quad \tilde{l}_1 = \begin{pmatrix} -1 \\ x \\ 0 \end{pmatrix} \quad \tilde{l}_2 = \begin{pmatrix} -1 \\ x \\ 0 \end{pmatrix}$$

$$[t_3]_x = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \tilde{E}_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \tilde{l}_1 = \begin{pmatrix} y \\ -x \\ 0 \end{pmatrix} \quad \tilde{l}_2 = \begin{pmatrix} y \\ -x \\ 0 \end{pmatrix}$$

for t_1 and t_2 , $\tilde{l}_1$ and $\tilde{l}_2$ are parallel, epipole is at infinity

$$\tilde{e}_1 = \begin{pmatrix} -y \\ 0 \end{pmatrix} \times \begin{pmatrix} c \\ -d \end{pmatrix} = \begin{pmatrix} 0 \\ bc-ad \end{pmatrix} \quad \tilde{e}_2 = \begin{pmatrix} 0 \\ bc-ad \end{pmatrix}$$

b) base line: the optical axis of the two cameras same location as principal point

$$c) \tilde{x}_2^T \tilde{E} \tilde{x}_1 = \tilde{x}_2^T K_2^{-T} \tilde{E} K_1^{-1} \tilde{x}_1 = \tilde{x}_2^T \tilde{E} \tilde{x}_1 = 0$$

$$\tilde{x}_2 = \tilde{x}_2 \quad \tilde{x}_1 = \tilde{x}_1 \quad K_1 = K_2 = I$$

Triangulation

$$P_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad P_2 = \begin{pmatrix} -2 & 0 & 1 & -3 \\ 0 & -2 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

$$A_1 = \begin{bmatrix} x_1^s P_{13}^T - P_{11}^T \\ y_1^s P_{13}^T - P_{12}^T \end{bmatrix} = \begin{pmatrix} -1 & 0 & \frac{1}{4} & 0 \\ 0 & -1 & \frac{1}{2} & 0 \end{pmatrix} \quad A_2 = \begin{bmatrix} x_2^s P_{23}^T - P_{21}^T \\ y_2^s P_{23}^T - P_{22}^T \end{bmatrix} = \begin{pmatrix} 2 & 0 & -\frac{1}{4} & \frac{14}{5} \\ 0 & 2 & -\frac{1}{4} & -\frac{1}{5} \end{pmatrix}$$

$$A \tilde{x}_w = 0 \quad \tilde{x}_w = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$$

Stereo Vision

$$a) z = \frac{fB}{d} \quad \Delta z = \frac{fB}{d-\Delta d} - \frac{fB}{d} = \frac{fB\Delta d}{d(d-\Delta d)} \approx \frac{fB\Delta d}{d^2} = \frac{z^2 \Delta d}{d^2}$$

b) increase b